

A healthy regression set is the **minimal** set of queries capturing the **maximal** spread of capability signatures.

We curate it continuously — **admit · drop · swap · review** each incoming query under a hard budget.

To our knowledge, no published industrial pipeline addresses this platform-side curation problem.

The regression-economics paradox

- Every onboarding customer ships an eval set tuned to their domain.
- The platform's **living regression set** must live under a **hard query-count ceiling**.
- It is **LLM-judged on every candidate release build** and **human-triaged on every failure** — gating flights, A/B tests & release certification.
- Admit-all inflates the engineering tail; reject-freely opens production blind spots.

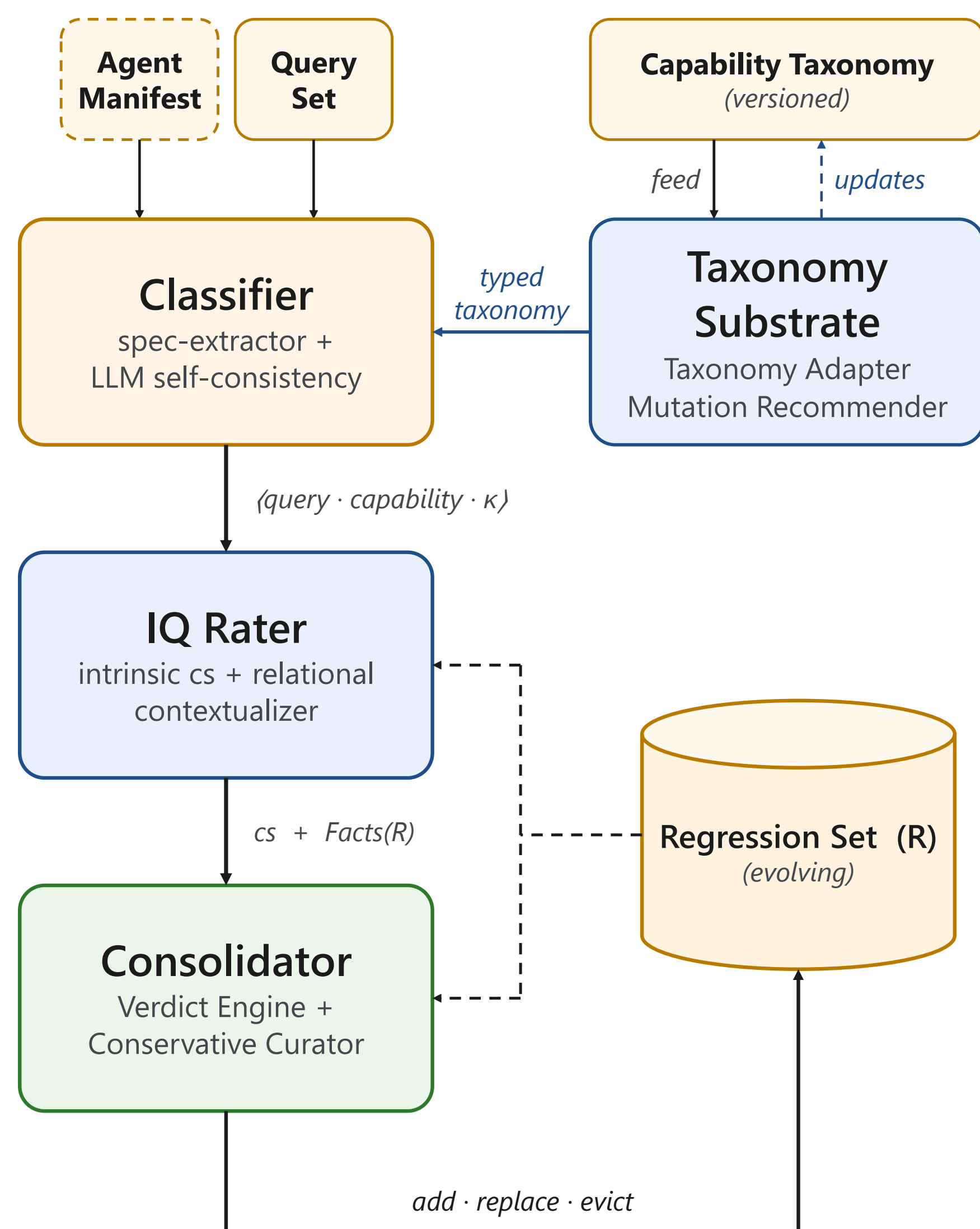
Two questions per incoming query

- Q1** Which capabilities does it exercise?
Q2 Given the live set, does it deserve a slot?

recurring cost per ADMITTED query



Pipeline at a glance



Why real queries, not synthetic?

Every slot is expensive (above), so each must reflect **how customers actually build and phrase**.

- Combinatorial blowup:** ~30 typed capabilities → intractable joint space; uniform sampling invents configurations no customer ships.
- Phrasing-sensitive LLM behavior:** Real failures ride on real wording, a distribution synthetic paraphrases cannot reproduce.

Capability taxonomy & signatures

Taxonomy for Copilot Extensibility (~30 typed leaves, versioned)

Request

- Auth** {None, API-Key, OAuth, SSO}
- Attachments** {Image, File, Audio}
- Localization** {Locale, RTL, Multilingual}
- ...

Execution

- Orchestration** {single/multi-turn, parallel, branching}
- Tool-Output Size** {S, M, L}
- Error Handling** {Retry, Fallback, Surface}
- ...

Response

- Format** {Plain, Structured, URLs, Tables}
- Citations** {None, Inline, Grounded}
- Tone** {Neutral, Formal, Concise}
- ...

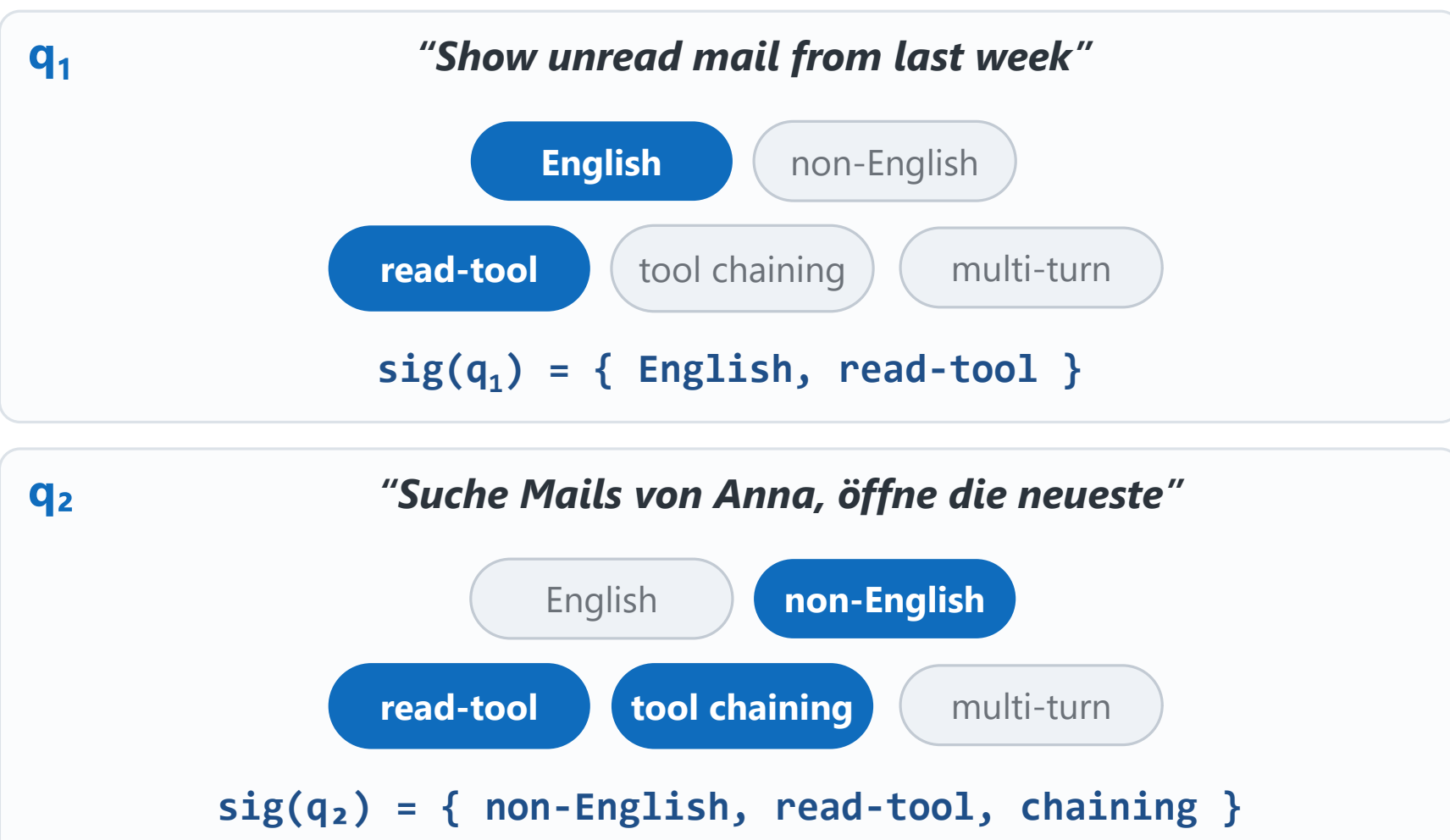
$\text{sig}(q) = \{c: \text{platform capability is exercised by the query}\}$ — the **capability signature** of a query.

Goal: Keep R small while $\bigcup \text{sig}(q)$ stays wide. The taxonomy is versioned — off-distribution clusters surface as extension candidates.

Example: One agent, two queries → two capability signatures

Agent: Mailbox assistant

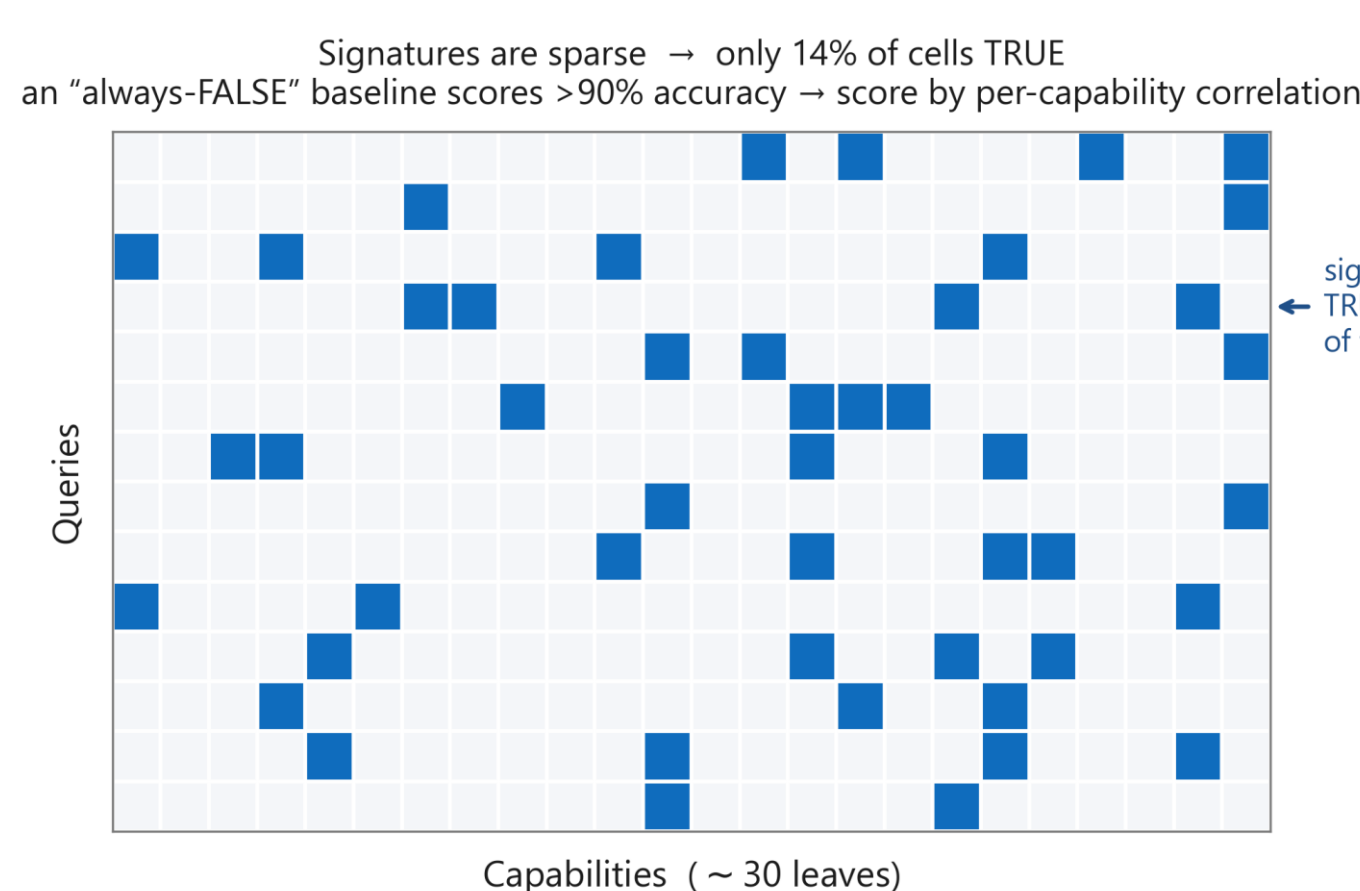
Tools: read_mail(filter, since) · search_mail(query)



① Classifier

Emits verdict v and confidence $\kappa \in [0, 1]$ per (query, capability) via two non-overlapping paths:

- Spec-based extractor:**
 - Deterministic lookup for declared capabilities (auth, tools).
 - Authoritative where the spec speaks ($\kappa = 1$)
- LLM, N-sample self-consistency:**
 - For intent-only capabilities (format, branching, tonality).
 - Majority vote ($\kappa = \text{mean exercise-probability}$)



② Invocation Quality (IQ) rater

Question Answered: When a new query shares a signature with an incumbent, is it a **better** test?

Principle: A shallow tester should yield to a deeper one.

- Adapter (intrinsic, stored):** a fact about q alone — persisted across cycles.
- Contextualizer (relational, live):** q -vs- R facts, recomputed every cycle because R changes on every edit.

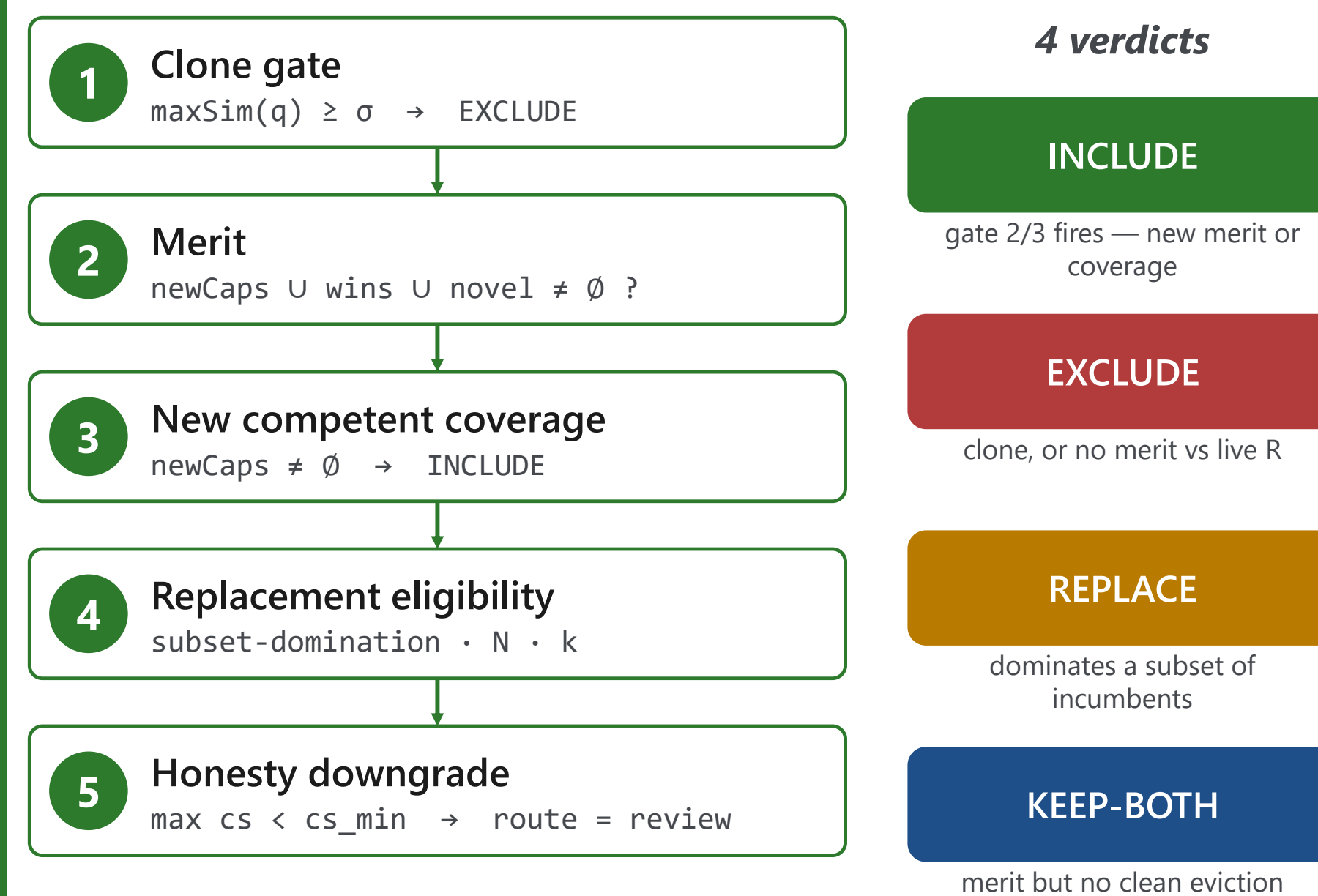
The math the IQ rater emits — all arg max / max / $\bigcup$ range over $r \in R$

Computation	What it answers
$\text{cs}(q, c) = \kappa$	How thoroughly q exercises c — stored
$\text{champion}(c) = \text{argmax}_r \text{cs}(r, c)$	Incumbent best test of c
$\text{champ_cs}(c) = \max_r \text{cs}(r, c)$	Current bar to beat on c
$\text{capst}(q) = \{c: \text{cs}(q, c) \geq \tau\}$	q 's competently-covered caps
$\text{newCaps}(q) = \text{capst}(q) \setminus \bigcup_r \text{capst}(r)$	q fills a coverage gap
$\text{wins}(q) = \{c: \text{cs}(q, c) > \text{champ_cs}(c) + \delta\}$	q beats the champion (margin δ)
$\text{maxSim}(q) = \max_r \text{sim}(q, r)$	Near-duplicate / clone signal
$\text{novel}(q) = [\text{sig}(q) \notin \{\text{sig}(r)\}]$	Unseen capability combination

③ Consolidator: verdict cascade

A pure rule-based **Verdict Engine** fires five gates in fixed order

Each contributes at most one reason code.



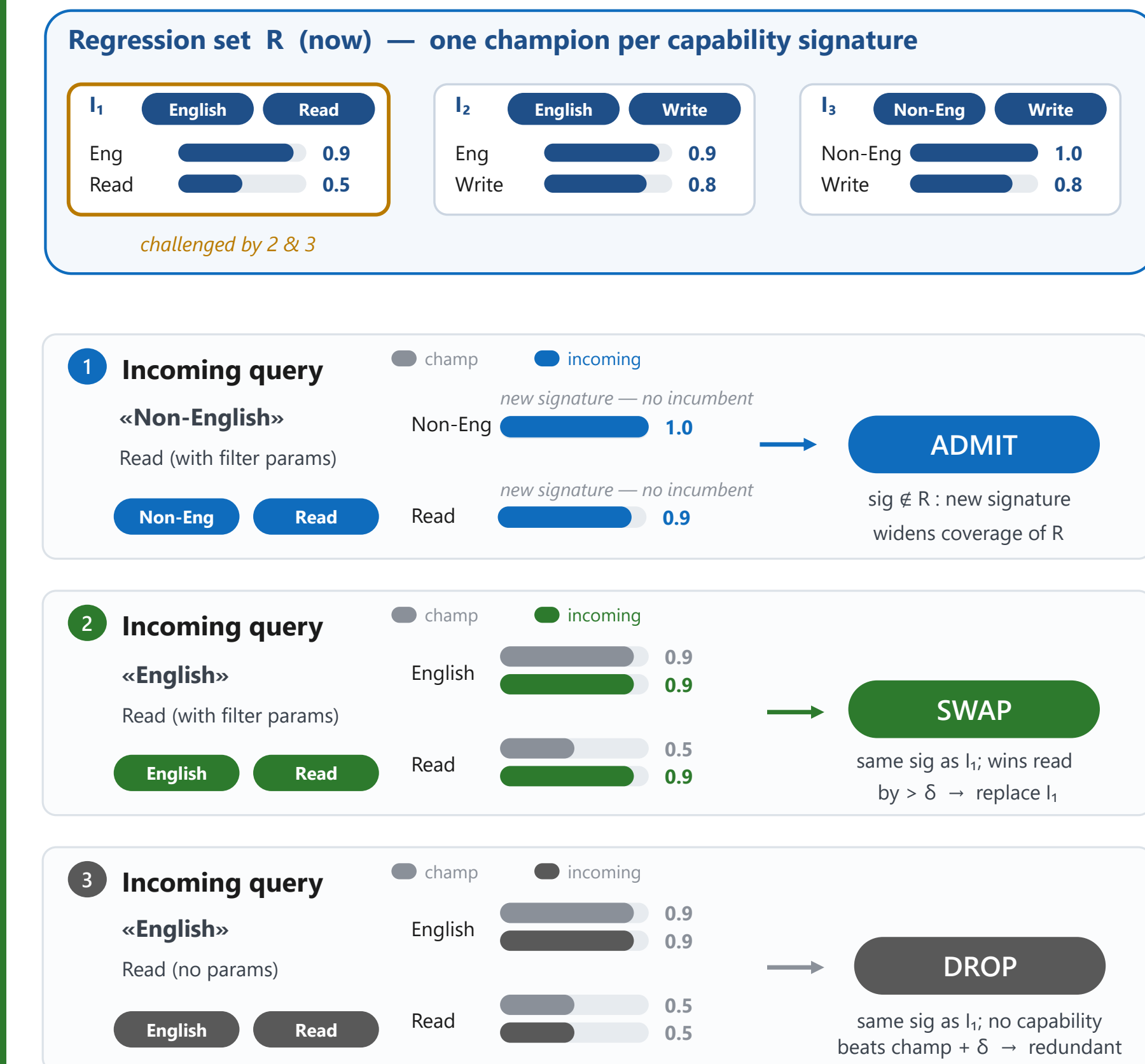
Curator — the only writer to R.

Never auto-deletes:

Beaten residents become **suggest-only** eviction items a human confirms. Admission cheap, eviction slow & auditable.

Worked example

A new query must earn its slot — scored per capability, not 0/1



Discussion — what generalizes

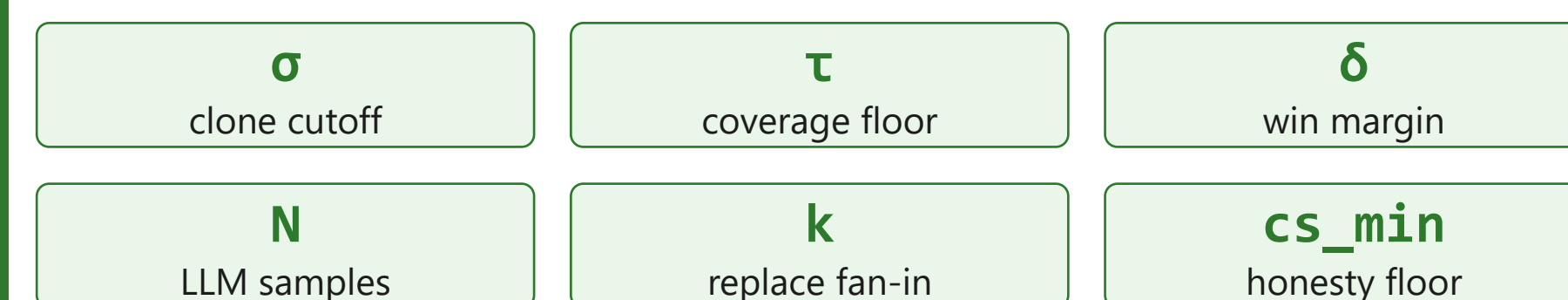
Generalizes beyond Microsoft 365 Extensibility

- Taxonomy-agnostic** — classifier, rater & consolidator assume only a typed taxonomy with exactly-one groups; nothing is M365-specific.
- Applies wherever a platform team owns a taxonomy, contributors ship eval sets, and the curated set is living & budget-bounded.
- The taxonomy itself evolves** — off-distribution query clusters become extension candidates a reviewer promotes to the next version (Taxonomy Adapter + Mutation Recommender).

Per-capability calibration — the central tuning problem

- The six knobs are **absolute** thresholds, not batch percentiles — the bar for auth $\neq$ the bar for response-format.
- Target: a **per-capability schedule** learned from validated reviewer overrides. Today: one uniform conservative default from a pilot.

Six configuration knobs — versioned, never in code



Accessible as an MCP server

- One tool per step (**classify · rate · consolidate**) plus step-ranges (classify→verdict, cs→verdict).
- Any MCP-aware agent invokes it — reviewers triage from chat; partners embed it into onboarding without re-implementing.